

Lab 1:

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × 

dvdrental/postgres@postgres

Query Query History

```

1  SELECT
2      relname AS object_name,
3      pg_size_pretty(pg_table_size(fed.oid)) AS data_size,
4      pg_size_pretty(pg_indexes_size(fed.oid)) AS index_size,
5      pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
6  FROM pg_class fed
7  JOIN pg_namespace n
8      ON n.oid = fed.relnamespace
9  WHERE fed.relkind = 'r'
10     AND n.nspname = 'public'
11  ORDER BY pg_total_relation_size(fed.oid) DESC;

```

Data Output Messages Notifications

Showing rows: 1 to 16

	object_name name	data_size text	index_size text	total_size text
1	rental	1232 kB	1120 kB	2352 kB
2	payment	896 kB	920 kB	1816 kB
3	film	736 kB	200 kB	936 kB
4	film_actor	272 kB	216 kB	488 kB
5	inventory	232 kB	208 kB	440 kB
6	customer	96 kB	112 kB	208 kB
7	address	88 kB	64 kB	152 kB
8	film_category	72 kB	40 kB	112 kB
9	city	64 kB	48 kB	112 kB
10	actor	40 kB	32 kB	72 kB
Total rows: 16 Query complete 00:00:00.649				
11	store	8192 bytes	32 kB	40 kB
12	staff	16 kB	16 kB	32 kB
13	language	8192 bytes	16 kB	24 kB
14	country	8192 bytes	16 kB	24 kB
15	category	8192 bytes	16 kB	24 kB
16	secure_orders	8192 bytes	16 kB	24 kB

“payment”, “customer”, “store”, “language”, “country”, “category”, “secure_orders” are “Index Heavy”

Lab 2:

Step 1: Check current size of 'payment' indexes

[Query](#) [Query History](#)

1 `SELECT pg_size_pretty ( pg_indexes_size ( 'payment ' ) )`

[Data Output](#) [Messages](#) [Notifications](#)



	pg_size_pretty text
1	920 kB

Step 2: Create a heavy index on payment_date

[Query](#) [Query History](#)

1 `CREATE INDEX idx_payment_date_long ON payment (payment_date , amount,`
2 `customer_id ) ;`

[Data Output](#) [Messages](#) [Notifications](#)

CREATE INDEX

Query returned successfully in 74 msec.

Step 3: Check size again

)

[Query](#) [Query History](#)

1 `SELECT pg_size_pretty ( pg_indexes_size ( 'payment ' ) )`

[Data Output](#) [Messages](#) [Notifications](#)



Showing rows: 1 to 1   Page No: 1 of 1    

	pg_size_pretty text
1	1488 kB

New index bytes=After bytes–Before bytes = 1488 - 920 = 568 (kB)

Number of pages= Index size in bytes / 8kb = 71 (pages)

Lab 3:

Query

Query History

1

-- Bước 1: Kiểm tra có phim nào có replacement_cost > 25.00 không

2

SELECT *

3

FROM film

4

WHERE replacement_cost > 25.00;

Data Output

Messages

Notifications

Showing rows: 1 to 236

Page No: 1

of 1

	film_id [PK] integer	title character varying (255)	description text	release_year integer	language_id smallint
1	4	Affair Prejudice	A Fanciful Documentary of a Frisbee And a Lumberjack who must Chase a Monkey in A Shark Tank	2006	
2	7	Airplane Sierra	A Touching Saga of a Hunter And a Butler who must Discover a Butler in A Jet Boat	2006	
3	18	Alter Victory	A Thoughtful Drama of a Composer And a Feminist who must Meet a Secret Agent in The Canadian Rockies	2006	
4	30	Anything Savannah	A Epic Story of a Pastry Chef And a Woman who must Chase a Feminist in An Abandoned Fun House	2006	
5	34	Arabia Dogma	A Touching Epistle of a Madman And a Mad Cow who must Defeat a Student in Nigeria	2006	
6	37	Arizona Bang	A Brilliant Panorama of a Mad Scientist And a Mad Cow who must Meet a Pioneer in A Monastery	2006	
7	38	Ark Ridgemont	A Beautiful Yarn of a Pioneer And a Monkey who must Pursue a Explorer in The Sahara Desert	2006	
8	265	Dying Maker	A Intrepid Tale of a Boat And a Monkey who must Kill a Cat in California	2006	
9	52	Ballroom Mockingbird	A Thrilling Documentary of a Composer And a Monkey who must Find a Feminist in California	2006	
10	53	Bang Kwai	A Epic Drama of a Madman And a Cat who must Face a A Shark in An Abandoned Amusement Park	2006	
11	55	Barbarella Streetcar	A Awe-Inspiring Story of a Feminist And a Cat who must Conquer a Dog in A Monastery	2006	
12	61	Beauty Grease	A Fast-Paced Display of a Composer And a Moose who must Sink a Robot in An Abandoned Mine Shaft	2006	
13	64	Beethoven Exorcist	A Epic Display of a Pioneer And a Student who must Challenge a Butler in The Gulf of Mexico	2006	
14	66	Beneath Rush	A Astounding Panorama of a Man And a Monkey who must Discover a Man in The First Manned Space Station	2006	
15	68	Betrayed Rear	A Emotional Character Study of a Boat And a Pioneer who must Find a Explorer in A Shark Tank	2006	
16	70	Bikini Borrowers	A Astounding Drama of a Astronaut And a Cat who must Discover a Woman in The First Manned Space Station	2006	

Total rows: 236

Query complete 00:00:00.211

CRLF

Ln 4, Col 32

Create Partial Index

Query Query History	
1	CREATE INDEX idx_expensive_films
2	ON film (replacement_cost)
3	WHERE replacement_cost > 25.00;
Data Output Messages Notifications	
CREATE INDEX	
Query returned successfully in 88 msec.	

Find the size of this specific index

Query Query History

1

SELECT

2

indexrelname,

3

pg_size_pretty(pg_relation_size(indexrelid))

4

FROM pg_stat_user_indexes

5

WHERE indexrelname = 'idx_expensive_films';

Data Output Messages Notifications

<

Tạo full index

Query Query History

```
1 CREATE INDEX idx_replacement_cost_full
2 ON film (replacement_cost);
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 68 msec.

Query Query History

```
1 SELECT
2     indexrelname,
3     pg_size_pretty(pg_relation_size(indexrelid))
4 FROM pg_stat_user_indexes
5 WHERE indexrelname = 'idx_replacement_cost_full';
```

Data Output Messages Notifications

									SQL
	indexrelname		pg_size_pretty						
	name		text						
1	idx_replacement_cost_f...		40 kB						

Partial index : 16kB < Full index: 40kB